LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel system, general design requirements, pressure and gravity system and description of a typical basic fuel system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel system, general design requirements, pressure and gravity system and description of a typical basic fuel system.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system diagrams; Safety: Ensure no fuel vapors present; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft fuel system?
(b) Name two common methods used to deliver fuel to the engine.
(c) Why is fuel system reliability critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General design requirements for fuel systems • Safety features in fuel system design • Material selection for fuel tanks and lines • Corrosion prevention measures in fuel systems • Standard layout of fuel system components	L & Ds	PPT, IB	15 min
(b)	• Gravity feed fuel system operation • Pressure feed fuel system operation • Comparison of gravity and pressure systems • Fuel pump types and locations • Fuel filter and strainer functions	L & Ds	PPT, PDF	20 min
(c)	• Description of a typical basic fuel system • Fuel tank venting and overflow systems • Fuel quantity indication principles • Fuel transfer and scavenge systems • Emergency fuel supply procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the key difference between a gravity feed and a pressure feed fuel system.
(b) Describe the role of the fuel filter in a typical basic fuel system.
(c) What safety feature prevents fuel tank overfilling in a basic system?

12. Home Assignment and Follow Up	Prepare a brief note comparing the gravity and pressure fuel systems, highlighting their advantages and limitations, to be used as a bridge for understanding the hydraulic system in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Hydraulic system, general design requirements, typical layout of a basic hydraulic system and its brief description.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min